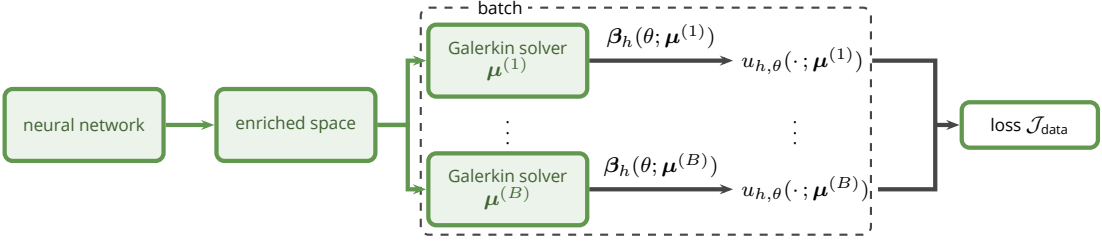
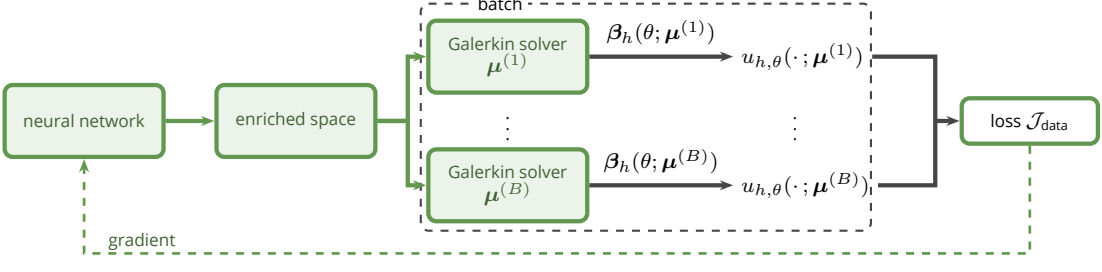
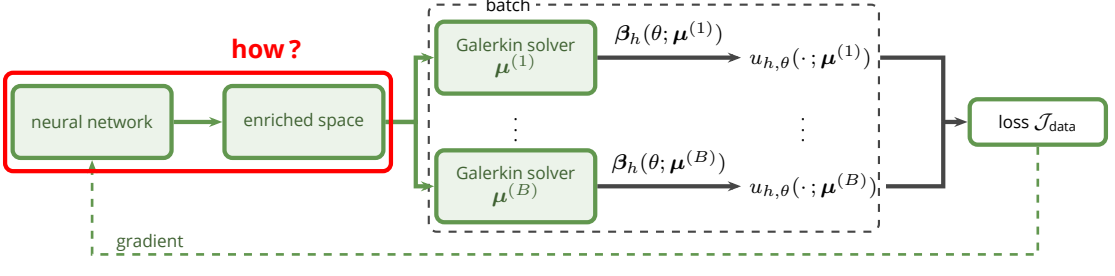

batch
neural network
enriched space
Galerkin solver
$\boldsymbol{\mu}^{(1)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(1)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(1)})$
Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(B)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(B)})$


batch
neural network
enriched space
Galerkin solver
$\boldsymbol{\mu}^{(1)}$
Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(1)})$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(B)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(1)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(B)})$
loss $\mathcal{J}_{\text{data}}$


batch
neural network
enriched space
Galerkin solver
$\boldsymbol{\mu}^{(1)}$
Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(1)})$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(B)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(1)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(B)})$
loss $\mathcal{J}_{\text{data}}$
gradient


how ?
neural network
enriched space
batch
Galerkin solver
$\boldsymbol{\mu}^{(1)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(1)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(1)})$
Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta;\boldsymbol{\mu}^{(B)})$
$u_{h,\theta}(\cdot\,;\boldsymbol{\mu}^{(B)})$
loss $\mathcal{J}_{\text{data}}$
gradient